QUBO Formulation: Planar Robot Inverse Kinematics Candidate Selection

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Problem Statement. Given a set of N discretized joint-angle candidates for a planar robot, the task is to select exactly one candidate such that the sum of the inverse kinematics (IK) error and a regularization cost associated with the selected candidate is minimized.

Decision Variables. Let N be the number of discretized joint-angle candidates. We introduce binary decision variables $x_i \in \{0, 1\}$ for each candidate index $i \in \{0, 1, \ldots, N-1\}$. The variable x_i indicates the selection of candidate i, where $x_i = 1$ if candidate i is chosen and $x_i = 0$ otherwise.

QUBO Derivation. Step 1: Objective Function. Let c_i denote the combined IK error and regularization cost for candidate i. The objective is to minimize the total cost:

$$\min \sum_{i=0}^{N-1} c_i x_i. \tag{1}$$

Since the QUBO model minimizes the Hamiltonian $H(x)$, we identify the objective contribution as:

$$H_{\text{obj}}(x) = \sum_{i=0}^{N-1} c_i x_i. \tag{2}$$

Step 2: Constraints. The problem requires that exactly one candidate be selected. This is enforced by the equality constraint:

$$\sum_{i=0}^{N-1} x_i = 1. \tag{3}$$

Step 3: Penalty Terms. We convert the constraint into a penalty term using a positive weight λ. The penalty Hamiltonian is:

$$H_{\text{pen}}(x) = \lambda \left(\sum_{i=0}^{N-1} x_i - 1 \right)^2. \tag{4}$$

Expanding the square yields:

$$H_{\text{pen}}(x) = \lambda \left(\sum_{i=0}^{N-1} x_i^2 - 2 \sum_{0 \le i < j \le N-1} x_i x_j + 1 \right). \tag{5}$$

Using the idempotent property of binary variables, $x_i^2 = x_i$, we simplify the quadratic sum of variables:

$$\sum_{i=0}^{N-1} x_i^2 = \sum_{i=0}^{N-1} x_i. \tag{6}$$

Substituting this back into the penalty expression gives:

$$H_{\text{pen}}(x) = \lambda \left(\sum_{i=0}^{N-1} x_i - 2 \sum_{0 \le i < j \le N-1} x_i x_j + 1 \right). \tag{7}$$

Distributing λ and separating terms, we obtain the general penalty contribution:

$$H_{\text{pen}}(x) = -\lambda \sum_{i=0}^{N-1} x_i + 2\lambda \sum_{0 \le i < j \le N-1} x_i x_j + \lambda. \tag{8}$$

Step 4: Combined Hamiltonian. The total QUBO Hamiltonian $H(x)$ is the sum of the objective and penalty terms:

$$H(x) = H_{\text{obj}}(x) + H_{\text{pen}}(x) = \sum_{i=0}^{N-1} c_i x_i - \lambda \sum_{i=0}^{N-1} x_i + 2\lambda \sum_{0 \le i < j \le N-1} x_i x_j + \lambda. \tag{9}$$

Grouping the linear terms, we write:

$$H(x) = \sum_{i=0}^{N-1} (c_i - \lambda) x_i + 2\lambda \sum_{0 \le i < j \le N-1} x_i x_j + \lambda. \tag{10}$$

Step 5: Q Matrix and Offset. The Hamiltonian is expressed in the standard QUBO form $H(x) = x^T Q x + c$, where $x = [x_0, \ldots, x_{N-1}]^T$. Reading off the coefficients from the general expression above, the entries of the symmetric matrix Q and the scalar offset c are:

$$Q_{i,i} = c_i - \lambda, \quad \forall i \in \{0, \ldots, N-1\}, \tag{11}$$

$$Q_{i,j} = 2\lambda, \quad \forall 0 \le i < j \le N-1, \tag{12}$$

$$Q_{j,i} = 2\lambda, \quad \forall 0 \le i < j \le N-1, \tag{13}$$

$$c = \lambda. \tag{14}$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any violation of the constraint results in a higher energy than any feasible solution. The maximum possible objective cost for a feasible solution is $\max_i c_i$. If the constraint is violated, the penalty term $\lambda(\sum x_i - 1)^2$ takes a value of at least λ (since $\sum x_i$ is an integer, any violation implies $|\sum x_i - 1| \ge 1$). Therefore, choosing $\lambda > \max_i c_i$ guarantees that the energy of any infeasible solution strictly exceeds the energy of any feasible solution, ensuring the global minimum corresponds to a feasible candidate.

Correctness Argument. Minimizing $H(x)$ over $x \in \{0,1\}^N$ recovers the optimal candidate selection. For any feasible solution satisfying $\sum_{i=0}^{N-1} x_i = 1$, the penalty term vanishes, and $H(x) = \sum_{i=0}^{N-1} c_i x_i = c_k$ where k is the unique index with $x_k = 1$. Thus, among feasible solutions, $H(x)$ equals the cost of the selected candidate. For infeasible solutions, the penalty term contributes at least λ. By the choice $\lambda > \max_i c_i$, the energy of any infeasible solution is strictly greater than the cost of the best feasible candidate. Consequently, the global minimum of $H(x)$ must be feasible and corresponds to the candidate with the minimum cost c_k.

Illustrative Example. Consider a concrete instance with $N = 3$ candidates and costs $c_0 = 10$, $c_1 = 5$, and $c_2 = 8$. We choose $\lambda = 100$, which satisfies $\lambda > \max\{10, 5, 8\}$. The QUBO Hamiltonian becomes:

$$\begin{aligned} H(x) &= (10 - 100)x_0 + (5 - 100)x_1 + (8 - 100)x_2 + 200x_0x_1 + 200x_0x_2 + 200x_1x_2 + 100 \\ &= -90x_0 - 95x_1 - 92x_2 + 200x_0x_1 + 200x_0x_2 + 200x_1x_2 + 100. \end{aligned} \tag{15}$$

The optimal solution is $x^* = [0, 1, 0]^T$, corresponding to selecting candidate 1. The minimum energy is $H(x^*) = -95 + 100 = 5$, which matches the cost c_1.